COMPUTER ENGINEERING

QUEZON CITY

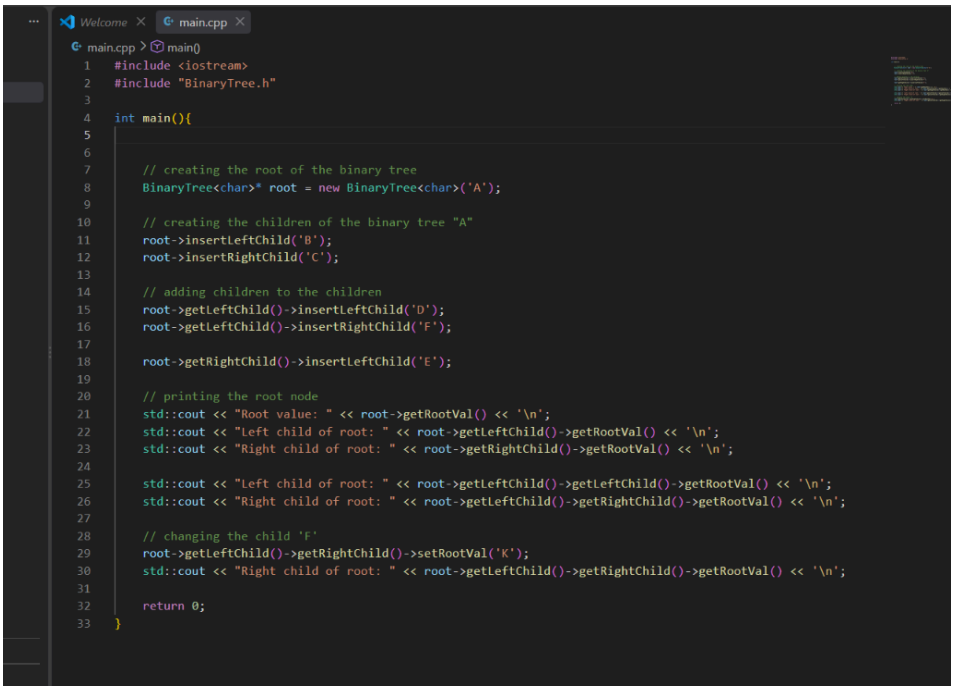
COURSE CODE/TITLE:	CPE 010 / Data Structure and Algorithm	**SECTION:**	CPE21S2
DATE PERFORMED:	09/16/2026	**DATE SUBMITTED:**	09/16/26
NAME:	Nieles, Cenndy M.	**INSTRUCTOR:**	Engr. Jimlord Quejado

ACTIVITY NO. 9.1 Seatwork Implementing Tress

1. PROCEDURE OUTPUT

What procedures did you perform?

Main.cpp



```cpp
#include <iostream>
#include "BinaryTree.h"

int main(){


    // creating the root of the binary tree
    BinaryTree<char>* root = new BinaryTree<char>('A');

    // creating the children of the binary tree "A"
    root->insertLeftChild('B');
    root->insertRightChild('C');

    // adding children to the children
    root->getLeftChild()->insertLeftChild('D');
    root->getLeftChild()->insertRightChild('F');

    root->getRightChild()->insertLeftChild('E');

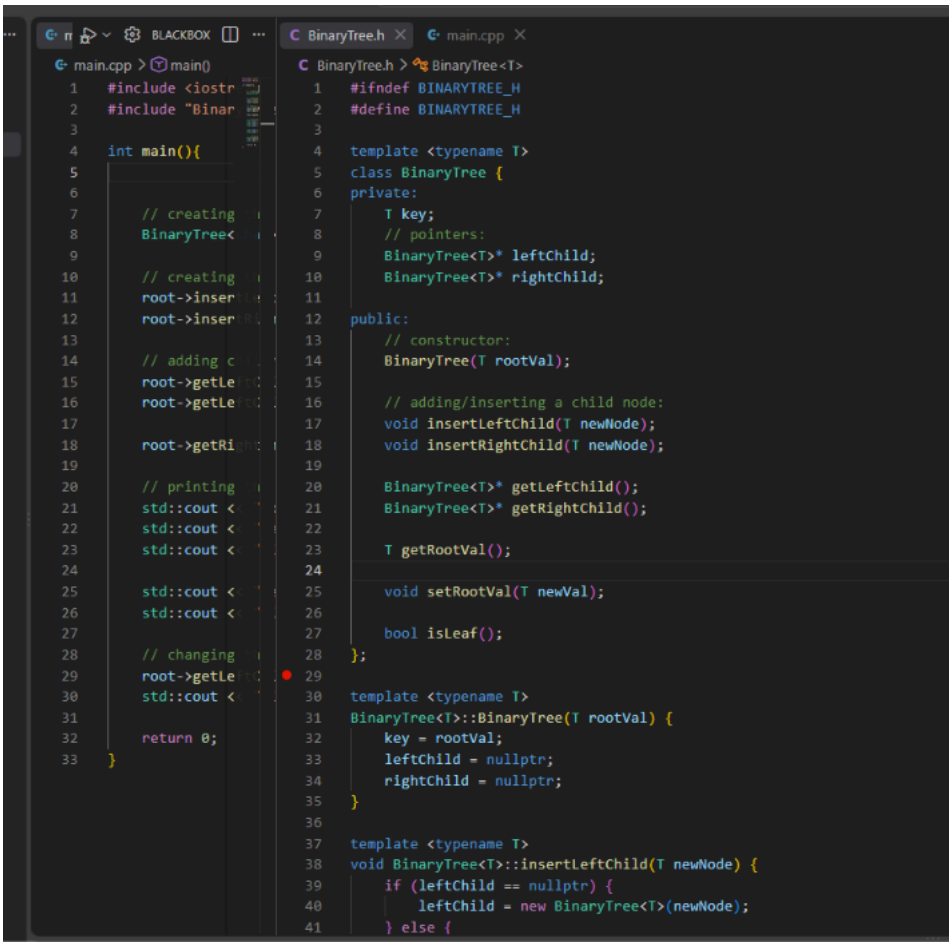
    // printing the root node
    std::cout << "Root value: " << root->getRootVal() << '\n';
    std::cout << "Left child of root: " << root->getLeftChild()->getRootVal() << '\n';
    std::cout << "Right child of root: " << root->getRightChild()->getRootVal() << '\n';

    std::cout << "Left child of root: " << root->getLeftChild()->getLeftChild()->getRootVal() << '\n';
    std::cout << "Right child of root: " << root->getLeftChild()->getRightChild()->getRootVal() << '\n';

    // changing the child 'F'
    root->getLeftChild()->getRightChild()->setRootVal('K');
    std::cout << "Right child of root: " << root->getLeftChild()->getRightChild()->getRootVal() << '\n';

    return 0;
}
```

BinaryTree.h



```cpp
#ifndef BINARYTREE_H
#define BINARYTREE_H

template <typename T>
class BinaryTree {
private:
    T key;
    // pointers:
    BinaryTree<T>* leftChild;
    BinaryTree<T>* rightChild;

public:
    // constructor:
    BinaryTree(T rootVal);

    // adding/inserting a child node:
    void insertLeftChild(T newNode);
    void insertRightChild(T newNode);

    BinaryTree<T>* getLeftChild();
    BinaryTree<T>* getRightChild();

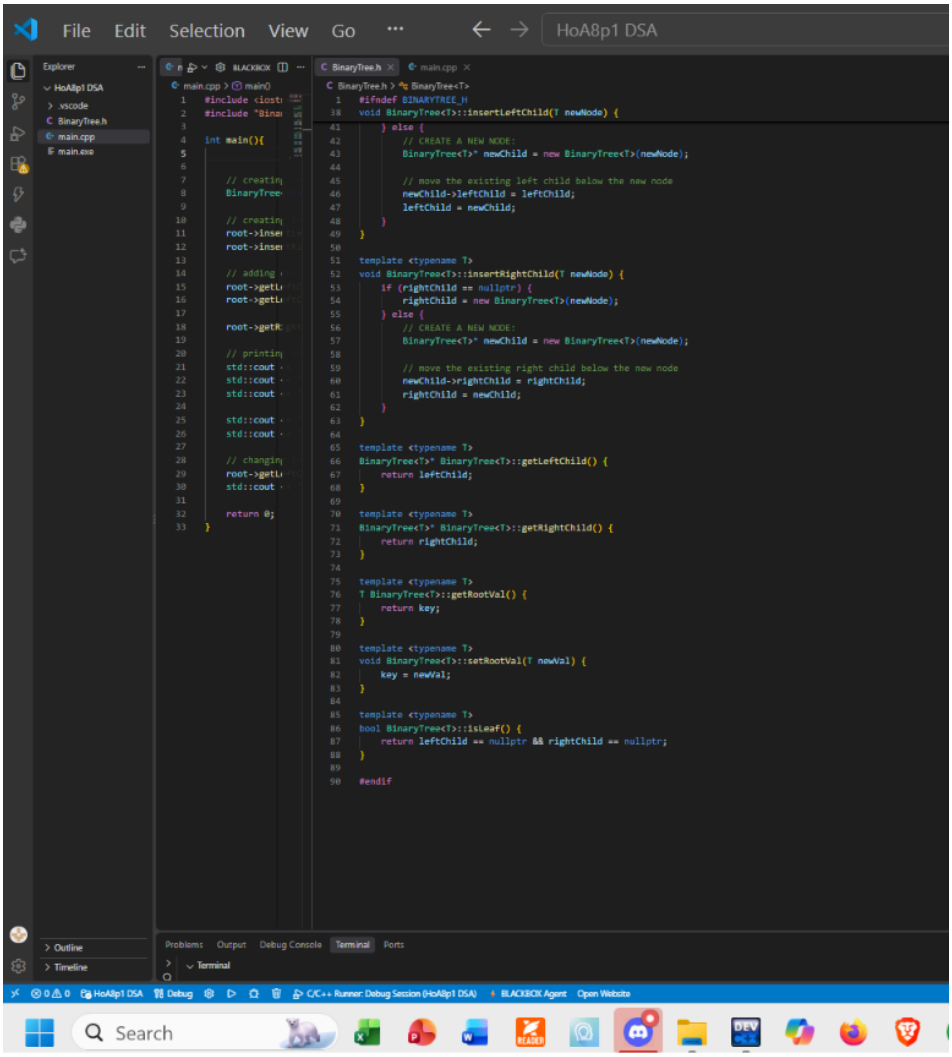
    T getRootVal();

    void setRootVal(T newVal);

    bool isLeaf();
};

template <typename T>
BinaryTree<T>::BinaryTree(T rootVal) {
    key = rootVal;
    leftChild = nullptr;
    rightChild = nullptr;
}

template <typename T>
void BinaryTree<T>::insertLeftChild(T newNode) {
    if (leftChild == nullptr) {
        leftChild = new BinaryTree<T>(newNode);
    } else {
```



Output

```
PS C:\Users\joyni\OneDrive\Desktop\HoA8p1 DSA> g++ main.cpp -o main
PS C:\Users\joyni\OneDrive\Desktop\HoA8p1 DSA> ./main
Root value: A
Left child of root: B
Right child of root: C
Left child of root: D
Right child of root: F
Right child of root: K
```

Analysis ;

Creates a Binary Tree Structure: It uses a template class (BinaryTree<T>) to build a dynamic tree where each node can hold data (like characters) and pointers to a leftChild and rightChild.Builds the Root and Initial Children: In main.cpp, it instantiates a root node with the character 'A' and gives it two direct children: 'B' (left) and 'C' (right). It navigates to node B and adds its own children: 'D' (left) and 'F' (right).It navigates to node C and adds 'E' as its left child. It navigates to node B and adds its own children ‘D’ in the left ‘F’ in the right and it navigates to node C and adds the E as its left Child. The print the node value it uses a getRootVal to traverse the branch and print out the values stored in the nodes such as printing the root A. it’s children B and C .Then modify Data it target a specific node using the pointer navigation and updates its value K using setRootVal then prints the updated value.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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